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THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'with apologies for the horrendous notation' and, in the next paragraph, 'then the the existence'; both are reproduced unchanged. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- K_0 of an additive or triangulated category by generators and relations. invoked at proof lines 36588-36590 and 36596-36598. Fixes the two groups being compared and makes both displayed maps well defined.
- Distinguished triangles from short exact sequences of projective modules, and from truncations of a bounded complex. used at proof lines 36597-36598 and 36604-36619. Is the tool that lets a statement about modules be promoted to a statement about complexes.
- The Euler characteristic map c of the preceding lemma and its additivity on distinguished triangles. invoked at statement line 36583. Supplies the map whose universal property is being established.

Source reference. Original lemma 0FCN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19188–19188. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 014Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1490–1490. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0118: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1354–1354. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 20288: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20288–20288. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 36491: sources/stacks-more-algebra/pinned-original-source.tex, source lines 36491–36491. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $D_{perf}(R)$ be the derived category of perfect objects, see Lemma 20288. The map c of Lemma 36491 gives an isomorphism $K_0(D_{perf}(R)) = K_0(R)$.*

Proof. It follows from the definition of $K_0(D_{perf}(R))$ (Derived Categories, Definition 0FCN) that c induces a homomorphism $K_0(D_{perf}(R)) \to K_0(R)$.

Given a finite projective module M over R let us denote $M[0]$ the perfect complex over R which has M sitting in degree 0 and zero in other degrees. Given a short exact sequence $0 \to M \to M' \to M'' \to 0$ of finite projective modules we obtain a distinguished triangle $M[0] \to M'[0] \to M''[0] \to M[1]$, see Derived Categories, Section 014Z. This shows that we obtain a map $K_0(R) \to K_0(D_{perf}(R))$ by sending $[M]$ to $[M[0]]$ with apologies for the horrendous notation.

It is clear that $K_0(R) \to K_0(D_{perf}(R)) \to K_0(R)$ is the identity. On the other hand, if $M^\bullet$ is a bounded complex of finite projective R-modules, then the the existence of the distinguished triangles of "stupid truncations" (see Homology, Section 0118)

$$\sigma_{\geq n} M^\bullet \to \sigma_{\geq n-1} M^\bullet \to M^{n-1}[-n+1] \to (\sigma_{\geq n} M^\bullet)[1]$$

and induction show that

$$[M^\bullet] = \sum (-1)^i [M^i[0]]$$

in $K_0(D_{perf}(R))$ (with again apologies for the notation). Hence the map $K_0(R) \to K_0(D_{perf}(R))$ is surjective which finishes the proof. □